

OCR Software

The benefits of OCR become clearer after you start using it. Though it's generally considered a tool for enterprises, home users and small offices can also benefit from it.

To start with, you can scan all your business cards, OCR them and maintain a searchable archive of them—you can even use special software to import scanned business cards.

Utilities called form-fillers enable you to scan printed forms and fill them electronically. Storing these forms electronically eliminates the need to rewrite the entire form each time a mistake is made.

Also, you can fill in multiple forms simultaneously—a feature that comes in handy for railway reservation forms.

How Does OCR Work?

OCR is used to scan and recognise text on paper, or convert scanned pages saved as images to text. It analyses the raster image and creates an index of areas that resemble possible text fields.

The software then attempts to recognise text characters by comparing the shape of the scanned objects to a database of words categorised by different fonts or typefaces.

Thereafter, it groups individual characters and compares them with the words in the dictionary that is set to use a particular language.

This step is extremely crucial for accuracy in recognition. The more comprehensive the dictionary, the more accurate is the finished product. The OCR software marks certain words that it 'considers' inaccurate for you to correct them manually.

Finally, the OCR software uses the index it created to align the text fields as accurately as possible.

OCR software accuracy depends on scanner quality, paper used to print text, etc. The latest breed of OCR packages uses optimisation algorithms, neural networks and even AI concepts to get this done. Using pattern-recognition techniques,

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 You Need	A cost-effective solution for home use
Look for	Support for documents with text and tables, the occasional newspaper and magazine clipping,
 You Need	A high-end software for professional usage
Look for	OCR support for fax, photocopies, typewritten documents, newspapers, magazines, etc

Net Spider
www.netspiderindia.com

the software tries to guess the character as a whole and look at all possibilities before arriving at a hypothesis. Some OCR packages have inbuilt tools that enable them to 'learn' from the changes you make to the output.

Field-specific recognition, wherein the scanned data is automatically stored in the appropriate field in, say, a database, will also be an integral part of the software in the near future.

Factors that affect OCR software accuracy

- ❑ Scanner quality
- ❑ Scan resolution
- ❑ Type of printed documents, whether laser printer outputs or photocopied
- ❑ Paper quality
- ❑ Fonts used in the text
- ❑ Linguistic complexities
- ❑ The dictionary used

Do Remember

A document that is free of wrinkles, ears, smudges or creases yield more accurate results. Keep a sheet of white paper behind a document while OCRing it, especially when scanning newspapers, thin paper, etc. This helps eliminating smudges during scanning.

Buying Tips

- ✓ Batch-processing comes in handy when you have a large number of documents to scan. With this feature you need not sit in front of the scanner feeding one page at a time. Batch-processing does need support from hardware as well; you should have a scanner that has an Automatic Document Feeder (ADF).
- ✓ Check for the languages supported by the OCR package. Although we are not expecting to see Hindi or any other regional language any time soon.

- ✓ Certain OCR software comes with the learning mode feature that allows it to 'learn' from the corrections you make in the recognized document.
- ✓ Apart from saving scanned files as Word documents, look for a Save As option that lets you save them in HTML, PDF and Excel formats as well.
- ✓ The OCR software should have a provision to add specific fonts and letter types into its dictionary, in case it fails to pick them up while scanning your documents.